# DNS Resolver Exposure Assessment Report

Team A3-PS012-TC045 · AIORI Hackathon 3 · PS-012 · ResolverLens 0.2.0 · run **phase5-main** · started 2026-10-28T08:52:07+00:00 · 20 authorised lab servers · 80 probes in 70.7 s

## 1. Summary

**8 of 20** servers provide recursion to the external client: t01 (bind), t02 (bind), t03 (unbound), t04 (unbound), t05 (dnsmasq), t14 (dnsmasq), t15 (unbound), t16 (bind). Every decisive verdict matches the seeded ground truth. Accuracy 100.0% (20/20), accuracy on decided cases 100.0%, coverage 80.0% (16 of 20 servers decided; 4 correctly reported as Inconclusive). On the question "does it provide recursion?", a single RA-flag check gets 20/20 right and ResolverLens 20/20 (section 10).




Recursive (8) Authoritative-only (4) Refused (4) Inconclusive (4)

How to read the labels. **Recursive**: the server looked up a name for the external client (proved by our own canary server's log). **Authoritative-only**: it answers its own zone but does not recurse. **Refused**: it denies the external client. **Inconclusive**: the evidence is missing or contradictory, and the reason is recorded. Verdicts follow the rules on the method page (section 9).

## 2. Findings per server

| Server | Address | Software | Verdict | Reason | Canary evidence (who asked the canary) | Ground truth (lab only) | Match |
|---|---|---|---|---|---|---|---|
| t01 | 10.10.0.11 | bind | **Recursive** | - | 10.10.0.11 | Recursive | **yes** |
| t02 | 10.10.0.12 | bind | **Recursive** | - | 10.10.0.12 | Recursive | **yes** |
| t03 | 10.10.0.13 | unbound | **Recursive** | - | 10.10.0.13 | Recursive | **yes** |
| t04 | 10.10.0.14 | unbound | **Recursive** | - | 10.10.0.14 | Recursive | **yes** |
| t05 | 10.10.0.15 | dnsmasq | **Recursive** | - | 10.10.0.15 | Recursive | **yes** |
| t06 | 10.10.0.16 | bind | **Refused** | - | - | Refused | **yes** |
| t07 | 10.10.0.17 | bind | **Refused** | - | - | Refused | **yes** |
| t08 | 10.10.0.18 | unbound | **Refused** | - | - | Refused | **yes** |
| t09 | 10.10.0.19 | unbound | **Refused** | - | - | Refused | **yes** |
| t10 | 10.10.0.20 | bind | **Authoritative-only** | - | - | Authoritative-only | **yes** |
| t11 | 10.10.0.21 | bind | **Authoritative-only** | - | - | Authoritative-only | **yes** |
| t12 | 10.10.0.22 | nsd | **Authoritative-only** | - | - | Authoritative-only | **yes** |
| t13 | 10.10.0.23 | nsd | **Authoritative-only** | - | - | Authoritative-only | **yes** |
| t14 | 10.10.0.24 | dnsmasq | **Recursive** | - | 10.10.0.11 | Recursive | **yes** |
| t15 | 10.10.0.25 | unbound | **Recursive** | - | 10.10.0.13 | Recursive | **yes** |
| t16 | 10.10.0.26 | bind | **Recursive** | - | 10.10.0.15 | Recursive | **yes** |
| t17 | 10.10.0.27 | unbound | **Inconclusive** | TIMEOUT | - | Inconclusive | **yes** |
| t18 | 10.10.0.28 | bind | **Inconclusive** | TIMEOUT | - | Inconclusive | **yes** |
| t19 | 10.10.0.29 | dnsmasq | **Inconclusive** | TIMEOUT | - | Inconclusive | **yes** |
| t20 | 10.10.0.30 | bind | **Inconclusive** | TIMEOUT | - | Inconclusive | **yes** |

The ground-truth column exists only because this is a seeded lab. A real assessment has no such column.

## 3. Evaluation against the seeded ground truth

Rows are the expected labels, columns are what ResolverLens reported. Green cells are correct, red cells are errors.

| Expected \ Reported | Recursive | Authoritative-only | Refused | Inconclusive | Total |
|---|---|---|---|---|---|
| **Recursive** | 8 | 0 | 0 | 0 | 8 |
| **Authoritative-only** | 0 | 4 | 0 | 0 | 4 |
| **Refused** | 0 | 0 | 4 | 0 | 4 |
| **Inconclusive** | 0 | 0 | 0 | 4 | 4 |

| Class | Precision | Recall | F1 | Expected | Reported |
|---|---|---|---|---|---|
| Recursive | 1.00 | 1.00 | 1.00 | 8 | 8 |
| Authoritative-only | 1.00 | 1.00 | 1.00 | 4 | 4 |
| Refused | 1.00 | 1.00 | 1.00 | 4 | 4 |
| Inconclusive | 1.00 | 1.00 | 1.00 | 4 | 4 |

| Measure | Value | Meaning |
|---|---|---|
| Accuracy | 100.0% | Verdicts equal to the expected label, over all servers |
| Accuracy on decided cases | 100.0% | Correct verdicts among servers where the tool gave a decisive label |
| Coverage | 80.0% | Share of servers the tool decided instead of reporting Inconclusive |
| Wrong decisions | 0 | Decisive verdicts that disagree with the expected label (the costly error) |
| False exposure / missed exposure | 0 / 0 | Reported Recursive when it should not be / failed to report Recursive when it should |
| Correct abstentions | 4 | Inconclusive reported for servers seeded as unusable |

## 4. False and inconclusive result analysis

### 4.1 False results

> No false results in this run: all 20 verdicts equal their expected labels. Each false result would be listed here with a likely cause (ACL, caching, forwarder chain, packet loss, rate limiting, upstream failure).

### 4.2 Inconclusive results

| Server | Software | Reason | Likely cause | Why the tool did not guess | Seeded as (lab only) | Expected? |
|---|---|---|---|---|---|---|
| t17 | unbound | TIMEOUT | packet loss, rate limiting or filtering | No reply to any canary attempt. A silent drop looks the same as packet loss, so the tool does not guess. | slow | **yes** |
| t18 | bind | TIMEOUT | packet loss, rate limiting or filtering | No reply to any canary attempt. A silent drop looks the same as packet loss, so the tool does not guess. | rate limited | **yes** |
| t19 | dnsmasq | TIMEOUT | packet loss, rate limiting or filtering | No reply to any canary attempt. A silent drop looks the same as packet loss, so the tool does not guess. | silent | **yes** |
| t20 | bind | TIMEOUT | packet loss, rate limiting or filtering | No reply to any canary attempt. A silent drop looks the same as packet loss, so the tool does not guess. | broken upstream | **yes** |

## 5. Exposure and remediation

RFC 5358 asks operators to offer recursion only to their intended clients. The following servers answered an external client:

| Server | Software | Suggested change (check the syntax for your version) |
|---|---|---|
| t01 (10.10.0.11) | bind | `allow-recursion { <your clients>; localhost; }; allow-query-cache { <your clients>; localhost; };` |
| t02 (10.10.0.12) | bind | `allow-recursion { <your clients>; localhost; }; allow-query-cache { <your clients>; localhost; };` |
| t03 (10.10.0.13) | unbound | `access-control: 0.0.0.0/0 refuse then access-control: <your clients> allow` |
| t04 (10.10.0.14) | unbound | `access-control: 0.0.0.0/0 refuse then access-control: <your clients> allow` |
| t05 (10.10.0.15) | dnsmasq | `local-service (or interface=<internal interface> with bind-interfaces)` |
| t14 (10.10.0.24) | dnsmasq | `local-service (or interface=<internal interface> with bind-interfaces)` Its lookups came from 10.10.0.11: that upstream server needs the same restriction. |
| t15 (10.10.0.25) | unbound | `access-control: 0.0.0.0/0 refuse then access-control: <your clients> allow` Its lookups came from 10.10.0.13: that upstream server needs the same restriction. |
| t16 (10.10.0.26) | bind | `allow-recursion { <your clients>; localhost; }; allow-query-cache { <your clients>; localhost; };` Its lookups came from 10.10.0.15: that upstream server needs the same restriction. |

## 6. Safety and compliance, measured from the evidence

| Check | Observed | Limit or rule |
|---|---|---|
| Destinations contacted | 20 (all on the allowlist: yes) | at most 20 authorised lab servers |
| Query types sent | A | type A only, never ANY |
| Peak rate, all targets together | 3 queries in any second | 5.0 per second plus burst |
| Peak rate, one target | 3 queries in any second | 1.0 per second plus burst |
| Largest reply / reply-to-query ratio | 251 bytes / 6.28 | 1232 bytes / 10.0 |
| How the run ended | completed | safety tripwire and stop file available |

The tool has no spoofing option: every query leaves from the probe client's own address. The lab network has no Internet access.

## 7. Evidence per server

### t01 · 10.10.0.11 · bind · Recursive

3/3 canary attempts: NOERROR, RA, marker answer, canary log hit (rule R1)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | NOERROR | 1 | 0 | - | - | 4.4 |
| canary 1: rl-77d8391459.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.11 | 2.4 |
| canary 2: rl-d866234a14.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.11 | 1.8 |
| canary 3: rl-1b2d48404e.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.11 | 1.3 |

### t02 · 10.10.0.12 · bind · Recursive

3/3 canary attempts: NOERROR, RA, marker answer, canary log hit (rule R1)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | NOERROR | 1 | 0 | - | - | 3.5 |
| canary 1: rl-07bcbd20d8.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.12 | 2.1 |
| canary 2: rl-83e6d8ea77.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.12 | 1.8 |
| canary 3: rl-a0ce6dbf3b.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.12 | 6.5 |

### t03 · 10.10.0.13 · unbound · Recursive

3/3 canary attempts: NOERROR, RA, marker answer, canary log hit (rule R1)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | NXDOMAIN | 1 | 1 | - | - | 1.2 |
| canary 1: rl-97feb347bb.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.13 | 2.6 |
| canary 2: rl-a181596430.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.13 | 1.2 |
| canary 3: rl-990512d128.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.13 | 1.3 |

### t04 · 10.10.0.14 · unbound · Recursive

3/3 canary attempts: NOERROR, RA, marker answer, canary log hit (rule R1)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | NXDOMAIN | 1 | 1 | - | - | 1.4 |
| canary 1: rl-5b99fbe282.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.14 | 3.4 |
| canary 2: rl-0de618e2eb.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.14 | 1.6 |
| canary 3: rl-512f5a1c40.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.14 | 1.6 |

### t05 · 10.10.0.15 · dnsmasq · Recursive

3/3 canary attempts: NOERROR, RA, marker answer, canary log hit (rule R1)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | REFUSED | 1 | 0 | - | - | 1.0 |
| canary 1: rl-8db0664812.canary.lab | 1 | NOERROR | 1 | 1 | 192.0.2.53 | seen from 10.10.0.15 | 1.3 |
| canary 2: rl-c7a51e6dc3.canary.lab | 1 | NOERROR | 1 | 1 | 192.0.2.53 | seen from 10.10.0.15 | 1.2 |
| canary 3: rl-9ae91f0a22.canary.lab | 1 | NOERROR | 1 | 1 | 192.0.2.53 | seen from 10.10.0.15 | 1.4 |

### t06 · 10.10.0.16 · bind · Refused

REFUSED for the baseline and 3/3 canary attempts (rule R2)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | REFUSED | 0 | 0 | - | - | 1.4 |
| canary 1: rl-57e0febe08.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.7 |
| canary 2: rl-1f15bdc207.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.7 |
| canary 3: rl-99dd415b24.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.7 |

### t07 · 10.10.0.17 · bind · Refused

REFUSED for the baseline and 3/3 canary attempts (rule R2)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | REFUSED | 0 | 0 | - | - | 4.2 |
| canary 1: rl-1111eddac7.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 3.5 |
| canary 2: rl-cb91e8660f.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.7 |
| canary 3: rl-63ae96533a.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.6 |

### t08 · 10.10.0.18 · unbound · Refused

REFUSED for the baseline and 3/3 canary attempts (rule R2)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | REFUSED | 0 | 0 | - | - | 0.9 |
| canary 1: rl-6a02490210.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.4 |
| canary 2: rl-d182657e37.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.4 |
| canary 3: rl-34a192617b.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 1.0 |

### t09 · 10.10.0.19 · unbound · Refused

REFUSED for the baseline and 3/3 canary attempts (rule R2)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | REFUSED | 0 | 0 | - | - | 1.8 |
| canary 1: rl-8d90996209.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.8 |
| canary 2: rl-4b6c6e297e.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.8 |
| canary 3: rl-fba542937d.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.5 |

### t10 · 10.10.0.20 · bind · Authoritative-only

baseline answered with AA; 3/3 canary attempts got no answer and no canary hit (rule R3)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | NOERROR | 0 | 1 | 10.10.0.20 | - | 1.5 |
| canary 1: rl-1915e15fd9.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 1.8 |
| canary 2: rl-ccc0cfd75b.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.9 |
| canary 3: rl-a620543b1d.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.7 |

### t11 · 10.10.0.21 · bind · Authoritative-only

baseline answered with AA; 3/3 canary attempts got no answer and no canary hit (rule R3)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | NOERROR | 0 | 1 | 10.10.0.21 | - | 1.5 |
| canary 1: rl-da89271127.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.7 |
| canary 2: rl-b9bc27df42.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.7 |
| canary 3: rl-9bd61ba233.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.7 |

### t12 · 10.10.0.22 · nsd · Authoritative-only

baseline answered with AA; 3/3 canary attempts got no answer and no canary hit (rule R3)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | NOERROR | 0 | 1 | 10.10.0.22 | - | 102.6 |
| canary 1: rl-95041817f6.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.6 |
| canary 2: rl-39d5371755.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.8 |
| canary 3: rl-64024fb4a4.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.6 |

### t13 · 10.10.0.23 · nsd · Authoritative-only

baseline answered with AA; 3/3 canary attempts got no answer and no canary hit (rule R3)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | NOERROR | 0 | 1 | 10.10.0.23 | - | 105.0 |
| canary 1: rl-651b197b65.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.6 |
| canary 2: rl-b0230c4116.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.6 |
| canary 3: rl-41e73502d6.canary.lab | 1 | REFUSED | 0 | 0 | - | - | 0.6 |

### t14 · 10.10.0.24 · dnsmasq · Recursive

3/3 canary attempts: NOERROR, RA, marker answer, canary log hit (rule R1)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | NOERROR | 1 | 0 | - | - | 4.6 |
| canary 1: rl-f04b9fef99.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.11 | 2.1 |
| canary 2: rl-22c41a13a8.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.11 | 2.1 |
| canary 3: rl-dfc0be95e9.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.11 | 2.4 |

### t15 · 10.10.0.25 · unbound · Recursive

3/3 canary attempts: NOERROR, RA, marker answer, canary log hit (rule R1)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | NXDOMAIN | 1 | 1 | - | - | 1.2 |
| canary 1: rl-1161416e6b.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.13 | 1.5 |
| canary 2: rl-dc9fe53ea8.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.13 | 1.5 |
| canary 3: rl-4dee5eb611.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.13 | 1.4 |

### t16 · 10.10.0.26 · bind · Recursive

3/3 canary attempts: NOERROR, RA, marker answer, canary log hit (rule R1)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | NOERROR | 1 | 0 | - | - | 3.2 |
| canary 1: rl-7a0e95262f.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.15 | 1.8 |
| canary 2: rl-50058e4e2f.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.15 | 1.6 |
| canary 3: rl-8694f75414.canary.lab | 1 | NOERROR | 1 | 0 | 192.0.2.53 | seen from 10.10.0.15 | 1.5 |

### t17 · 10.10.0.27 · unbound · Inconclusive (TIMEOUT)

no reply to any of the 3 canary attempts (rule R4)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | TIMEOUT | 0 | 0 | - | - | - |
| canary 1: rl-8a45d82147.canary.lab | 1 | TIMEOUT | 0 | 0 | - | - | - |
| canary 2: rl-64b1ca8b08.canary.lab | 1 | TIMEOUT | 0 | 0 | - | - | - |
| canary 3: rl-6f9fc3d164.canary.lab | 1 | TIMEOUT | 0 | 0 | - | - | - |

### t18 · 10.10.0.28 · bind · Inconclusive (TIMEOUT)

no reply to any of the 3 canary attempts (rule R4)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | NOERROR | 1 | 0 | - | - | 3.6 |
| canary 1: rl-81166ca59e.canary.lab | 1 | TIMEOUT | 0 | 0 | - | - | - |
| canary 2: rl-ae7b71a6d4.canary.lab | 1 | TIMEOUT | 0 | 0 | - | - | - |
| canary 3: rl-dc90473a01.canary.lab | 1 | TIMEOUT | 0 | 0 | - | - | - |

### t19 · 10.10.0.29 · dnsmasq · Inconclusive (TIMEOUT)

no reply to any of the 3 canary attempts (rule R4)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | TIMEOUT | 0 | 0 | - | - | - |
| canary 1: rl-0fb0be39d5.canary.lab | 1 | TIMEOUT | 0 | 0 | - | - | - |
| canary 2: rl-ff591d951d.canary.lab | 1 | TIMEOUT | 0 | 0 | - | - | - |
| canary 3: rl-0f59aab53a.canary.lab | 1 | TIMEOUT | 0 | 0 | - | - | - |

### t20 · 10.10.0.30 · bind · Inconclusive (TIMEOUT)

no reply to any of the 3 canary attempts (rule R4)

| Query | RD | Status | RA | AA | Answer | Canary log | ms |
|---|---|---|---|---|---|---|---|
| baseline : ns.lab.test | 0 | NOERROR | 1 | 0 | - | - | 3.6 |
| canary 1: rl-f9fcf887b5.canary.lab | 1 | TIMEOUT | 0 | 0 | - | - | - |
| canary 2: rl-dc7c4d8117.canary.lab | 1 | TIMEOUT | 0 | 0 | - | - | - |
| canary 3: rl-80bdc9af87.canary.lab | 1 | TIMEOUT | 0 | 0 | - | - | - |

## 8. Reproducibility

| Item | Value |
|---|---|
| Run, seed, tool version | phase5-main, 1337, 0.2.0 |
| Config digest (settings and targets file) | `a79e23245d905be41ed801a58b2f76ea343f1974bbfe7dcc4078dfdcb49b0a54` |
| Timeout, attempts, EDNS buffer | 2.0 s, 3, 1232 bytes |
| Re-run | `make probe RUN=<name>` then `make report RUN=<name>`. The same seed and run id give the same canary names, and the verdicts are computed from evidence.jsonl alone. |

## 9. Method: the four rules

Each server receives one baseline query (RD off, a name in its own zone) and 3 canary queries (RD on, a fresh unique name each time, in a zone served only by our canary server). The canary server logs every query it receives. Rules are checked in order and the first match decides:

| Rule | Verdict | Condition |
|---|---|---|
| 1 | Recursive | Every canary attempt is NOERROR with RA set, the marker answer, and a hit in the canary log |
| 2 | Refused | The baseline and every canary attempt are REFUSED |
| 3 | Authoritative-only | The baseline is answered with AA; every canary attempt gets a reply with no answer, no RA and no canary hit |
| 4 | Inconclusive | Anything else, with a reason: TIMEOUT, SERVFAIL, TC_FAIL, CONFLICT, ERROR or NO_DATA |

## 10. Comparison with single-flag checks

The same evidence judged by simpler rules, on one question: does the server provide recursion to the external client? A single-flag check reads only the first RD=1 reply. That is not enough when a server can set RA and return an answer without ever recursing (a sinkhole, a DNS firewall, a local-data override). ResolverLens also requires the canary server's log to show the lookup.

| Method | What it reads | Correct | False exposure | Missed exposure | Servers called wrongly |
|---|---|---|---|---|---|
| **ResolverLens** | flags, answer and canary evidence over 3 attempts | 20/20 | 0 | 0 | none |
| **RA-flag check** | first RD=1 reply: NOERROR and RA set (what Nmap's dns-recursion script reads) | 20/20 | 0 | 0 | none |
| **RA + AA check** | first RD=1 reply: NOERROR, RA set, AA clear and an answer | 19/20 | 0 | 1 | t05 |

| Server | Seeded as (lab only) | Truly recursive? | ResolverLens | RA-flag check | RA + AA check |
|---|---|---|---|---|---|
| t05 | open recursive | yes | **recursive** | **recursive** | **not recursive** |